Validation

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mrgsolve model

Load your mrgsolve model

```
runno <- "106"
mod <- mread(here("wk11/homework/mrg-model", paste0(runno, ".cpp")))
```

```
Building 106_cpp ... done.
```

see model

```
see(mod)
```

```
Model file:  106.cpp
[ prob ]
  ECP8506 mrgsolve model for illustration
  This model requires mrgsolve >= 1.0.3

[ pkmodel ] cmt = "GUT,CENT,PERIPH", depot = TRUE

[ param ]
  WT   = 70
  EGFR = 90
  ALB  = 4.5
  AGE  = 35

[ nmext ]
  path = "../nm-model/106/106.ext"
  root = "cppfile"

[ pk ]
  double V2WT   = log(WT/70.0);
  double CLWT   = log(WT/70.0)*0.75;
  double CLEGFR = log(EGFR/90.0)*THETA(6);
  double CLAGE  = log(AGE/35.0)*THETA(7);
  double V3WT   = log(WT/70.0);
  double QWT    = log(WT/70.0)*0.75;
  double CLALB  = log(ALB/4.5)*THETA(8);

  double KA  = exp(THETA(1) + ETA(1));
  double V2  = exp(THETA(2) + V2WT + ETA(2));
  double CL  = exp(THETA(3) + CLWT + CLEGFR + CLAGE + CLALB + ETA(3));
  double V3  = exp(THETA(4) + V3WT);
  double Q   = exp(THETA(5) + QWT);

  double S2 = V2/1000.0; //; dose in mcg, conc in mcg/mL

[ error ]
  double IPRED = CENT/S2;
```

```
  double Y = IPRED * (1+EPS(1));

[ capture ] CL V2 IPRED Y WT EGFR ALB AGE
```

Prepare input

Load your NONMEM input dataset

```
csv <- read_csv(here("wk11/homework/data/analysis3.csv"), na = ".")
```

```
Rows: 4360 Columns: 34
-- Column specification --------------------------------------------------------
Delimiter: ","
chr  (4): USUBJID, STUDY, ACTARM, RF
dbl (29): NUM, ID, TIME, SEQ, CMT, EVID, AMT, DV, AGE, WT, HT, EGFR, ALB, BM...
lgl  (1): C

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
head(csv, n = 3)
```

```
# A tibble: 3 x 34
  C       NUM    ID  TIME   SEQ   CMT  EVID   AMT    DV   AGE    WT    HT  EGFR
  <lgl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
1 NA        1     1  0        0     1     1     5   0    28.0  55.2  160.  114.
2 NA        2     1  0.61     1     2     0    NA  61.0  28.0  55.2  160.  114.
3 NA        3     1  1.15     1     2     0    NA  91.0  28.0  55.2  160.  114.
# i 21 more variables: ALB <dbl>, BMI <dbl>, SEX <dbl>, AAG <dbl>, SCR <dbl>,
#   AST <dbl>, ALT <dbl>, CP <dbl>, TAFD <dbl>, TAD <dbl>, LDOS <dbl>,
#   MDV <dbl>, BLQ <dbl>, PHASE <dbl>, STUDYN <dbl>, DOSE <dbl>, SUBJ <dbl>,
#   USUBJID <chr>, STUDY <chr>, ACTARM <chr>, RF <chr>
```

Load NONMEM output table contains PRED

The PRED and IPRED column in this table is generated by NONMEM. Let's rename it as NMPRED and NMIPRED to avoid confusion.

```
tab <- read_table(here("wk11/homework/nm-model", runno,
                       paste0(runno, ".tab")),
                  skip = 1) %>% rename(NMPRED=PRED, NMIPRED=IPRED)
```

```
-- Column specification --------------------------------------------------------
cols(
  NUM = col_double(),
  IPRED = col_double(),
  NPDE = col_double(),
  CWRES = col_double(),
  DV = col_double(),
  PRED = col_double(),
  RES = col_double(),
  WRES = col_double()
)
```

```
head(tab, n = 3)
```

```
# A tibble: 3 x 8
    NUM NMIPRED   NPDE  CWRES    DV NMPRED    RES   WRES
  <dbl>   <dbl>  <dbl>  <dbl> <dbl>  <dbl>  <dbl>  <dbl>
1     1     0    0      0       0      0    0      0
2     2    68.5 -0.633 -0.512  61.0   60.6  0.425 -0.533
3     3    90.8  0.403  0.126  91.0   78.5 12.4    0.143
```

Join NONMEM input dataset and output table

The NUM column provides a key to join these two

```
data <- left_join(tab, csv, by = "NUM")
head(data, n = 3)
```

```
# A tibble: 3 x 41
    NUM NMIPRED   NPDE  CWRES  DV.x NMPRED    RES   WRES C        ID  TIME   SEQ
  <dbl>   <dbl>  <dbl>  <dbl> <dbl>  <dbl>  <dbl>  <dbl> <lgl> <dbl> <dbl> <dbl>
1     1     0    0      0       0      0    0      0     NA        1  0        0
2     2    68.5 -0.633 -0.512  61.0   60.6  0.425 -0.533 NA        1  0.61     1
3     3    90.8  0.403  0.126  91.0   78.5 12.4    0.143 NA        1  1.15     1
```

```
# i 29 more variables: CMT <dbl>, EVID <dbl>, AMT <dbl>, DV.y <dbl>, AGE <dbl>,
#   WT <dbl>, HT <dbl>, EGFR <dbl>, ALB <dbl>, BMI <dbl>, SEX <dbl>, AAG <dbl>,
#   SCR <dbl>, AST <dbl>, ALT <dbl>, CP <dbl>, TAFD <dbl>, TAD <dbl>,
#   LDOS <dbl>, MDV <dbl>, BLQ <dbl>, PHASE <dbl>, STUDYN <dbl>, DOSE <dbl>,
#   SUBJ <dbl>, USUBJID <chr>, STUDY <chr>, ACTARM <chr>, RF <chr>
```

Validation using `PRED`

Simulate `PRED` using `mrgsolve`

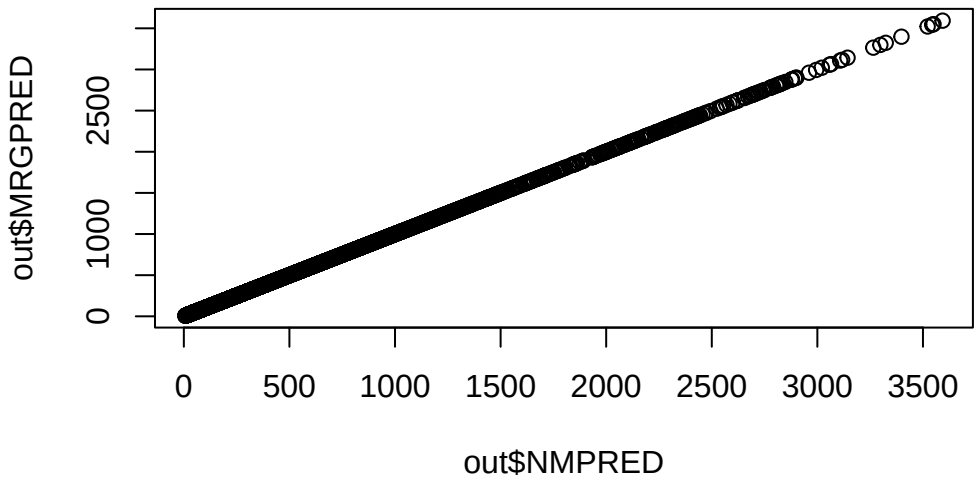
```
out <-
  mod %>%
  # Remove all random effects, so simulated IPRED=PRED
  zero_re() %>%
  # Take observation records only
  obsonly() %>%
  mrgsim(data = data,
         # Recover `NMPRED` for validation
         recover = "NMPRED",
         # Specify the output format as dataframe (df)
         output = "df",
         # Limit the number of output digits to 5 (what `NONMEM` is giving).
         digits = 5) %>%
  rename(MRGPRED=IPRED) # Rename mrgsolve `PRED` as `MRGPRED`

head(out, n = 3)
```

```
  ID TIME     GUT   CENT   PERIPH     CL     V2 MRGPRED      Y    WT   EGFR ALB
1  1 0.61 1.93350 2.9359 0.064809 3.0306 48.463  60.580 60.580 55.16 114.45 4.4
2  1 1.15 0.83380 3.8064 0.177380 3.0306 48.463  78.543 78.543 55.16 114.45 4.4
3  1 1.73 0.33785 4.0241 0.312260 3.0306 48.463  83.035 83.035 55.16 114.45 4.4
    AGE NMPRED
1 28.03 60.580
2 28.03 78.543
3 28.03 83.035
```

Graphical check PRED by NONMEM versus mrgsolve

```
plot(out$NMPRED, out$MRGPRED)
```



Numerical check PRED by NONMEM versus mrgsolve

```
summary(100*(out$NMPRED - out$MRGPRED)/out$NMPRED)
```

```
      Min.    1st Qu.     Median       Mean    3rd Qu.       Max.
-0.0087843  0.0000000  0.0000000  0.0002597  0.0000000  0.0097656
```

Validation using IPRED

Read in an additional table that contain the post-hoc ETA values

Validation using IPRED values needs an additional table which has post-hoc ETA values.

```
etas <- read_table(here("wk11/homework/nm-model", runno,
                        paste0(runno, "par.tab")),
                   skip = 1) %>%
  dplyr::select(matches("NUM|ETA"))
```

```
-- Column specification --------------------------------------------------------
cols(
  NUM = col_double(),
  CL = col_double(),
  V2 = col_double(),
  Q = col_double(),
  V3 = col_double(),
  KA = col_double(),
  ETA1 = col_double(),
  ETA2 = col_double(),
  ETA3 = col_double()
)
```

Join the additional dataset with post-hoc ETA values with the `data` created earlier. Again, the `NUM` column provides a key to join these two

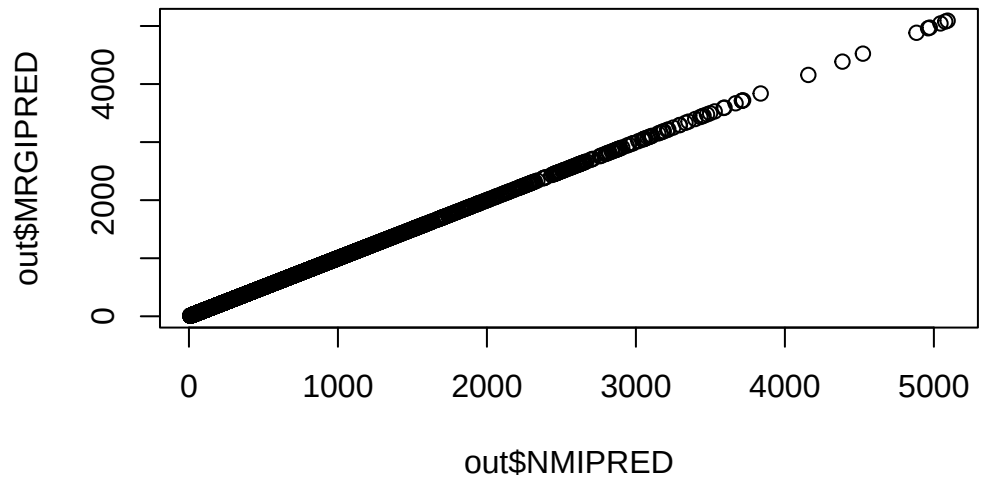
```
data <- left_join(data, etas, by = "NUM")
```

Simulate `IPRED` using `mrgsolve`

```
out <- mrgsim(
  mod,
  data,
  # Specify the source of ETA values coming from data
  etasrc = "data.all",
  # Take observation records only
  obsonly = TRUE,
  # Recover `NMPRED` for validation
  recover = "NMIPRED, NMPRED",
  # Specify the output format as dataframe (df)
  output = "df",
  # Limit the number of output digits to 5 (what `NONMEM` is giving).
  digits = 5
) %>% rename(MRGIPRED=IPRED)
```

Graphical check IPRED by NONMEM versus mrgsolve

```
plot(out$NMIPRED, out$MRGIPRED)
```



Numerical check PRED by NONMEM versus mrgsolve

```
summary(100*(out$NMIPRED-out$MRGIPRED)/out$NMIPRED)
```

```
      Min.    1st Qu.     Median       Mean    3rd Qu.       Max.
-0.0087268  0.0000000  0.0000000  0.0002906  0.0000000  0.0099522
```